

**INTELLIGENT AUTOMATED SEATING ALLOCATION SYSTEM FOR
LARGE SCALE ASSEMBLIES USING CONFIGURABLE CONSTRAINT
OPTIMIZATION**

DARIUS LEE SHIN

**A project report submitted in partial fulfilment of the
requirements for the award of the degree of
Bachelor of Computer Science (Hons)**

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Universiti Teknologi PETRONAS**

May 2026

DECLARATION

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at UTP or other institutions.

Signature :  _____

Name : DARIUS LEE SHIN

ID No. : 22003269

Date : 30 July 2026

APPROVAL FOR SUBMISSION

I certify that this project report entitled **“INTELLIGENT AUTOMATED SEATING ALLOCATION SYSTEM FOR LARGE SCALE ASSEMBLIES USING CONFIGURABLE CONSTRAINT OPTIMIZATION”** was prepared by **DARIUS LEE SHIN** has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Computer Science (Hons) at Universiti Teknologi PETRONAS.

Approved by,

Signature :  _____

Supervisor : Ts. Dr. Ooi Boon Yaik

Date : 30 July 2026

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INTELLIGENT AUTOMATED SEATING ALLOCATION SYSTEM FOR LARGE SCALE ASSEMBLIES USING CONFIGURABLE CONSTRAINT OPTIMIZATION

ABSTRACT

Seating allocation for large-scale assembly event is a multi-criteria combinatorial problem that becomes complex when participant profiles, venue layout restrictions, event specific contribution tiers, accessibility needs, and event-specific rules must be considered simultaneously by the event administrator. This final year project is titled as Intelligent Automated Seating Allocation System for Large-Scale Assemblies Using Configurable Constraint Optimization to address the current manual seating allocation workflow of PJKIT that create the issues of inconsistencies on seating plan, rework, and participant seat-retrieval bottlenecks. The project aims to design and develop a configurable constraint-based allocation engine, a controlled incremental reallocation mechanism for ad hoc participant changes, and a participant-facing seat lookup interface linked to the latest published plan. During the planning phase of the project, literature on seating allocation problem, constraint optimization, existing event seating systems, and CP-SAT solver was reviewed, and requirement engineering was conducted through stakeholder interviews, site observations, and a joint application development workshop with Netizen eXperience and PJKIT representatives. The proposed methodology models PJKIT seating rules as hard constraints and weighted soft constraints, that could be solve using Google OR-Tools CP-SAT with Python and deployed through AWS Lambda within the existing platform. Future work will cover the mathematical formulation, development and deployment of the solver function, integrate administrator and participant interfaces, and validate system using realistic event scenarios and performance metrics.

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LIST OF SYMBOLS / ABBREVIATIONS

<i>CP</i>	Constraint Programming
<i>COP</i>	Constraint Optimization Problem
<i>PJKIT</i>	Petaling Jaya Kwan Inn Teng
<i>NX</i>	Netizen eXperience
<i>SA</i>	Seating Allocation
<i>CP-SAT</i>	Constraint Programming – Satisfiability

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CHAPTER 1

INTRODUCTION

1.1 Background

Large-scale assembly events such as community gatherings, convocation, ceremonies, and religious assemblies frequently require event administrator to choose the seating for their event participants. Although the task of seating assignment for event participants appears to be an administrative task, it is a multi-criteria combinatorial assignment problem (Rossi et al., 2006; Ipsen et al., 2026). Each participant must be matched and assigned to exactly one seat in a limited seating layout, and each seating assignment must simultaneously adhere and satisfy a set of interrelated event-specific rules. For example, participants of specific categories may need to be placed in specific seating zones, participants with higher event priority or contribution may receive a more favourable seat, and certain seats may be blocked or reserved from event participants. The number of possible assignments grows explosively with scale, which even in the simplified case where each of the case study's 150 participants assigned a single seat among a venue that can support up to 232 seats, the number of possible ordered seat assignments is denoted as $\frac{232!}{(232-150)!}$ approximately of 8.75×10^{326} possible combinations for the seating plan arrangement, which is infeasible to check manually by human, and it is also impractical for machine to enumerated it naively (Rossi et al., 2006). The seating allocation task is also computationally hard, and not merely large. The structural rules that govern PJKIT seating, which are the two-seat Emperor allocation units, the contribution ordering within each tier, the demand-driven tier bands and the contiguous filling of rows, place this problem outside the class of

assignment problems that classical polynomial-time methods can solve (Ipsen et al., 2026).

The case-study organization of this project, which Petaling Jaya Kwan Inn Teng (PJKIT), organized large annual community events that consisted of 100 to 200 participants per event. These events are held in a hall that could be set up to 16 rows and 16 seats per row, which in total 232 assignable seats after structurally blocked divider in the centre of the assembly hall based on the given layout from previous events. The current seating assignments workflow implemented by PJKIT is a typical manual procedure across many event administrator, which participant records are first store in excel spreadsheets, then the event administrator create the seating plan by manually filling each participant's name in a predefined excel spreadsheet template as shown in Figure 1 below, the created seating plan will then be will convert into PDF document that will be shared digitally with participants through WhatsApp and used to print out an A2 sized venue chart.

2024年 新春法會功德主排位															
三寶佛															
西單1															東單1
西單2															東單2
西單3															東單3
西單4															東單4
西單5															東單5
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Figure 1.1 PJKIT's Annual Event Venue Seating Layout

The seating allocation decisions made by PJKIT event administrator are not arbitrary for every single event, instead they adhere to a set of event-specific rules that caused the seating allocation problem to be more complicated than general seat assignment that relies on selection and randomness. In the context of PJKIT annual event-specific rule, participants will be divided into three (3) contribution tiers, which the Emperor, Merit and Bodhi. The contribution tiering is used as the primary criteria to determine the precedence of the seating arrangements from the front of the hall, which each Emperor row should precede the first Merit Row, and every Merit row should precede the first Bodhi row. A further characteristic of PJKIT registration is that Emperor participants attend as couples, so one Emperor registration occupies two adjacent seats rather than one. The seating boundaries for each tier are not a fixed zone, but it is determined by the demand. On the other hand, each tier seating row also has their allocation rule to determine which participant will be allocated toward a more favourable seat in each row. For example, the participant with high contribution should be seated in a higher favourable row compared to the less contributed participant within the same tiering. The event administrator from PJKIT have to hold many competing considerations in mind at once to balance these coupled allocation rules manually for over a hundred of participants every event, which contribute to the inefficiency as the seating map preparation usually consumed more than a week from the starting of manually assigning seat, validate until finalize the seating plan.

Commercial event-planning tools such as Cvent, PerfectTablePlan, and WeddingWire offer features including floor plan design, table arrangement, guest list management, drag-and-drop seating and printable seating charts. While the WeddingWire Seating Chart Tools offer similar features for creating floor plans, assigning guests, and exporting or sharing arrangements (The Knot Worldwide Inc., 2026). PerfectTablePlan manages guest details, preferences, manual drag-and-drop or automatic seat assignment, it also supports printable charts (Oryx Digital Ltd., 2026). However, these solutions prioritize visualization of seat allocation, guest arrangement, and manual planning support rather than explicit fairness-driven allocation controlled by customizable weighted constraint.

The consequences that will be faced by PJKIT of continuing their current manual seating allocation workflow are severe and compounding. First, the quality of seating allocation decline as scale of participants increases, two different event staff member allocating the same event would result in two different plans, and neither of them could be objectively justified against the community's stated tier, pairing and priority rule, which causing the event administrator to be expose to the perceptions of unfairness. Then, the preparation of the seating map consumes an unreasonable amount of volunteer time in the weeks before the event. Any late change, such as an absence, a substitution, or a newly registered walk-in ad hoc participant, requires an error -prone rework on a nearly completed seating plan. Each manual revision and reallocation of participants seat may increase the risk of introducing duplicate seat assignments, or missing participants.

This final year project is carried out in collaboration with Netizen eXperience, a software company that provides end-to-end digital solutions in Southeast Asia, which operates a Corporate Social Responsibility (CSR) event management platform currently used by PJKIT to manage events and participants registrations and historical record. Netizen eXperience has appointed a Tech Lead as the industry supervisor to provide technical advice and facilities development for the seating allocation system. The industry supervisor also acts as the intermediary for requirement gathering, validation and organizing the site visit for discussion with PJKIT stakeholders. The seating allocation system proposed in this project is designed to be integrated as a new system module into the staging environment for the existing event management platform of Netizen eXperience rather than a stand-alone academic prototype.

1.2 Problem Statements

This project addresses three (3) pain points that are identified from the background of study in Section 1.1 and validated against the current event seating map generation workflow of PJKIT.

Problem Statement 1: Inconsistency and inefficiency in multi-criteria manual seating allocation workflow

The manual participant seating allocation workflow requires the event administrator to consider many participant profiles simultaneously, including participant's contribution, tiering priority, activeness on participating in other events organized by PJKIT, and status of participation against a constrained seating layout with demand-driven tier groups. This process is slow and does not scale with the growth of participants number and seating layout expansion of the event. The quality of the produced seating plan is difficult to validate or justify against the administrator's own stated event-specific rules, and different event administrators make different decisions on assigning seats to participants resulting in producing an inconsistent seating map.

Problem Statement 2: High operational disruption and lack of controlled dynamic reallocation mechanisms

Manual workflows provide limited support for a controlled reallocation of seating maps following changes to participant data closely to end of event preparation, and on the event day due to ad hoc events such as participant absence or substitution of a participant. Once a plan has been published, such a change creates a fundamentally different problem from initial allocation, it is a repair problem, in which the assignment that are directly affected by the changes must be fixed while preserving the rest of the previously communicated plan, that includes participants who may already have been informed or seated at their locations. A late change that appears in the existing manual participant seating allocation workflows may force event administrator from PJKIT to revise the whole seating plan, which increases the risk of duplicate or outdated seat records, and re-incurs the full cost and inconsistency of manual allocation for every small change.

Problem Statement 3: Communication bottlenecks and inefficiencies in participant seat retrieval

Participants from PJKIT events currently relies on several ways to retrieve their assigned seats details in an event including getting from the printed seating map that will be showing during the event day, seats listing PDF document that shared by PJKIT event administrator or by asking the event staff to check manually for them during the

event day. This administrator-driven distribution of participants seating information creates queues at the event, and it didn't provide assurance that participants are reading the latest approved plan and offers no-self-service channel for participants to check their assigned seat before or during the event.

1.3 Aims and Objectives

The objectives of the thesis are derived directly from the three (3) problem statements and specify what the project aimed at developing, solve and achieve which shown as following:

- i) **To design and develop a configurable, constraint-based seating allocation engine** that converts participant profiles, seating layouts and configurable event-specific rules into programmable constraints for automating the seating plan generation.
- ii) **To develop a controlled incremental dynamic seating reallocation mechanism** that accommodates ad hoc participant data changes due to absence, substitutions of participant and late registration, including on the event day by treating the latest published plan as the baseline state and modifying only the affected seats.
- iii) **To develop a participant-facing seat lookup interface** that allows participants to retrieve and view their assigned seat number and event seating map from the latest event administrator published seating map.

1.4 Scope of Project

The scope of this project consists of five (5) modules, the core is the development of the seat allocation engine and solver function, which is a service that takes in the participant's profile, seating layout, and the event rules, then returns a complete and

optimised seating plan as the solution, it is the principal technical and research contribution for this project. The project has defined four (4) more supporting modules that could complement with the seat allocation engine module. All the scope defined below represents the workload to be completed by December 2026.

1.4.1 Seat Allocation Engine and Solver Function Development

The primary scope is the design and development of the seat allocation engine. The engine will receives a structured input that will be validate with schema, which are the participant profiles, the seating layout, the constraint configuration, and the latest published seating map if existed, and it will generate a seating allocation result along with metric that could be used to indicate the status of the result generated from the solver function.

The event-specific rules of PJKIT are modelled as programmable mathematical constraint expressions, where hard constraints represent the mandatory rules that every seating map must satisfy and weighted soft constraints represent the preferences of the event administrator on seating allocation that must be optimized. The output of the engine includes the participant-to-seat assignments together with by minimizing the cost of violation of the soft-constraints, which are the solver status, the independent hard-constraint validation result, the total weighted soft-constraint penalty with a per-constraint breakdown, the movement count and distance where applicable, and the runtime. This module is the core research contribution of the project because it determines how a seating map can be generated systematically from participant profiles and modelled event rules from PJKIT.

1.4.2 Event Administrator Seat Allocation Management Module

The proposed system includes an event administrator facing seating allocation operation dashboard, which is the interface that allows the event administrator to view participant list retrieved from the Netizen eXperience Management Web App's

database, then select required participants for generating the seating plan with the deployed Seat Allocation Solver Function. Event Administrator can also optionally configure the constraint weightage for the seating allocation system before the generation of new seating plan with this dashboard. Upon the generation of the seating plan, Event Administrator can review the generated seating plan result in the dashboard, regenerating when necessary.

Versioning and publishing are important part of this module, where every generated seating plan result is treated as a version that the event administrator may compare and review, and only the approved version is made available to the public by publishing the seating plan in the dashboard. This module is included as part of the project, because seating allocation must remain under the control of the event administrator, as the system is intended to support the decision making during the seating allocation process rather than replace the human review that the PJKIT's event administrators currently perform for all the events.

1.4.3 Configurable Constraint-Based Allocation

The system should also support configurable constraint-based seating allocation, where the event-specific rules are represented and classified as hard constraint and weighted soft constraint before being processed and construct by the seating allocation engine with the solver function. The hard constraints include demand-driven tier grouping seating following the order of the event contribution tiering in PJKIT, and at most one occupant per seat. While the soft constraint of the system includes rules such as activeness of the participant on previous event in PJKIT, and movement minimization upon reallocation. The soft constraint weights carry no fixed ranking, and their weightage are configurable per event, which reflects the situational trade-offs described by the PJKIT stakeholders during elicitation activities. This module is playing an important role in ensuring that the seating allocation system generates rule-consistent and priority-aware seating maps rather than sequential assignment or

include randomness in seating allocation. The full constraint classification listing is mentioned in Section 3.5.3.

1.4.4 Dynamic Reallocation upon Participant Data Changes on ad hoc event

The system also supports dynamic reallocation as the controlled incremental repair of a published seating map following participant data changes due to ad hoc event such as the absence of a registered participant and the substitution of the participant seat on the event day. The dynamic reallocation is distinguished from the initial seating plan generation from scratch mentioned in Section 1.4.1, where the generation solves the seating allocation problem on every selected participant and every seat before any seating map exists, while the reallocation treats the published seating map as the baseline state and processes only the changed participant data.

The system classifies participants into two (2) categories before the execution of the reallocation which affected participant and unaffected participant. The affected participant category covers the absent participant, the replacement participant and any occupant of a candidate seat included in the reallocation, while the unaffected participant category refers to the participant that their assigned seat is preserved. A complete regeneration of the seating map remains available in the admin dashboard but only as a final solution that the event administrator must explicitly invoke to trigger the change to the entire seating plan during an event. Every repaired result is stored as a new version with a movement summary and required review before publication to the participant view.

1.4.5 Event Participant Seat Lookup Interface

The system supports a participant-facing seat lookup interface that enable the participant to view and check their assigned seat details such as number of the seat and refer to the full event seating map after the event administrator has published the

seating plan by triggering the generate seating plan function that mentioned in the Section 1.4.1. This module addresses the current dependence on printed seating maps, and PDF listing that shared multiple time through WhatsApp due to changes in participant data. The detailed authentication implementation is treated as a supporting function inherited from the host platform rather than built it from scratch.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Search Strategy

The literature and previous studies reviewed in this chapter was identified through a database searches conducted on reliable online academic research tools and platform including Google Scholar, IEEE Xplore, and the ACM Digital Library. The keywords that used were the combination of “*seating allocation*”, “*seat assignment*”, “*office space allocation*”, “*constraint optimization problem*”, “*constraint programming*”, and the “*CP-SAT Solver*”. The search was restricted to publications and previous study in English language. A previous research study and work was included as part of the literature review in this project, when it is related to either the solution for assigning people to physical seats or presented an algorithmic constraint optimization approach that is transferable to the problem.

For the review on existing systems and solution in the market that mentioned in Section 2.3 was conducted across several categories of solution that an event administrator in PJKIT could fairly adopt which including the manual practice, enterprise event management platform, dedicated seating planners with automatic assignment, consumer seating chart tools, and reserved-seat ticketing and venue access software. One (1) or two (2) representatives’ products were reviewed in each category, and every assessment is based on the vendor’s publicly available product documentation, feature pages and demo videos inspected in July 2026.

2.2 Academic Studies on Seating and Space Allocation

This section will be focusing a series of academic studies that addresses seating and spectator allocation problem as a combinatorial optimization problem. There are Four (4) existing studies' article are reviewed in this section.

First, Ipsen et al. (2026) introduced the Hierarchical Seating Allocation (HSA) problem that extends the generic Seating Allocation (SA) problem in which the seats on a commercial office must be distributed to teams within an organizational hierarchy, so that teams that are related hierarchically are seated near to each other in the office floor plan. They demonstrated that the HSA problem can be solve by decompose the problem into sub-problems of SA that could be solved with a top down approached. They compared four (4) SA sub-problem solvers, first is an Integer Program (IP) that they adapted from Hales and Garcia (2019), which used to selects a central seat for each team and minimize the distance from the team members to it in the office, followed by an Iterative Clustering Algorithm (ICA) which is a heuristic method that closely resembles k-means, also a regret-based greedy algorithm to form a baseline for the IP method, and a warm-started local neighbourhood search heuristics (LS) that re-optimizes the current solution.

The limitation of this study for this project is that its objective is to get the proximity of hierarchical related teams on an unstructured floor plan in a commercial office. It does not consider the scenario which the seating layout is a defined row and position geometry, also this study does not cover the multi-seat allocation unit, individual ordering, and an approach to repair a seating plan that have been published to the participant. The local search neighbourhood search heuristics (LS) mentioned in the study aimed to improves a solution within a single planning run but does not prove the ability of preserving the seats that has been published to the participant.

Hales and Garcia (2019) address the allocation problem of physical seats to parties in a legislative chamber using mathematical optimization that form the Integer Program Seating Allocation (IPSA) method presented by Ipsen et al. (2026). This research study is contextually the closet to the PJKIT business case study in this project,

because a legislative chamber shared the essential properties of a community assembly hall, which both are having a fixed geometry of rows and seats to act as the space of allocation, groups that are entitled to blocks of adjacent seating, and a ordering status that determines which blocks are the more desirable in the event. However, the overall study focus on the seating optimization is differs from the PJKIT seating allocation business use case, because the placement in the study is determined by group membership rather than focus on individual contribution amount and tiering in an event, and there is no ordering constraint among individuals with each group in the legislative chamber, which again differ from PJKIT use case that also prioritize the sequence of seating allocation for individual within the same contribution tiering.

Moreover, Muñoz et al. (2006) address ticket-to-seat allocation problem for a Formula 1 Grand Prix at a scale of up to 50,000 seats. The formulated ticket groups carry a category, a priority rank and a dispersion flag, while the seats carry a zone, a row, a category, a rank and a status. The rules are divided explicitly into mandatory rules, which a ticket group must be assigned seats of a matching category and status, and nine (9) optional rules which cover subgroup splitting, rank alignment between the ticket and the seat, the minimum and maximum number of tickets in a row, never leaving a single ticket isolated, avoiding empty seats at the edges of rows, and distributing occupancy evenly so that a partially sold venue still appears full. The implemented solver adapts the region's growing technique from computer vision image segmentation, in which a seed seat is first selected for each ticket subgroup and then it is grown outward by iteratively taking the best scoring neighboring seat, after which a local search pass improves the first candidate solution. The method is anytime method, which means that a usable solution appears within minutes and improves if more time is allowed. The relevant of this study to the project's PJKIT business use case is that they separate the mandatory rules and the optional rules independently, which matched the hard and soft constraint architecture adopted in this project.

To conclude, none of the reviewed studies provides the combined capability that the PJKIT case study requires. Ipsen et al. (2026) and Hales and García (2019) achieve exact optimization for the mentioned seating allocation problem, but through a custom formulation that are built for one venue and one rule set, with no configurable

rule layer and no mechanism for repairing a seating map that has already been published to participants. Muñoz et al. (2006) achieve scale and anytime behaviour, but with no optimality or completeness guarantee and no explicit solver status. However, none of the reviewed existing works or studies model a multi-seat allocation unit, a demand-driven tier band with a shareable boundary row, or a contribution ordering among individuals within the same tier, and none of them treats the repair of an already-published plan as a separate issue to be optimized.

2.3 Existing Seat and Space Allocation System

Table 2.1 lists the existing seat and space allocation systems reviewed across the five (5) categories defined in Section 2.1.

Table 2.1 Existing Seat and Space Allocation Systems

System	Main Capability	Limitation
Manual Excel Spreadsheet	Allows organizers to manually record participant names, seat numbers, groupings, and remarks.	No automated allocation, no fairness metric, no optimization, and difficulty handling dynamic changes.
Cvent <i>Cvent Inc. (2026)</i>	Supports event management, event diagramming, collaboration, and floor planning.	Strong in event planning and visualization but limited in explicit fairness-driven weighted constraint optimization.
PerfectTablePlan <i>Oryx Digital Ltd. (2026)</i>	Supports guest details, RSVPs, preferences, automatic assignment, drag-and-drop seating, last-minute changes, and printed floor plans.	Provides useful automatic seating support, but formal fairness metrics and configurable weighted constraint optimization are not the focus.

System	Main Capability	Limitation
WeddingWire Seating Chart Tool <i>The Knot Worldwide Inc. (2026)</i>	Supports drag-and-drop seating charts, RSVP tracking, table shape customization, sharing, printing, and exporting.	Useful for event seating visualization but limited in formal optimization and fairness evaluation.
Prisimm Floor Planning Software <i>Prisimm, Inc. (2026)</i>	Provides cloud-based floor plan design with 2D and 3D venue diagramming, guest list management, guest grouping and colour coding, and drag-and-drop placement of guests at tables within the designed floor plan. Captures guest requirements such as VIP placement requests and dietary needs.	Guests are assigned to tables rather than to numbered row and position seats. Placement is manual drag-and-drop and no automatic or optimized assignment is documented. Guest requirements are captured as annotations rather than as constraints that drive the allocation, and there is no guest-facing seat lookup.
Eventbrite Reserved Seating <i>Eventbrite. (2026, July 24)</i>	Can creates a custom venue seat map through drag-and-drop design and sells numbered seats through a clickable floor plan on which the attendee selects their own seat. The seat label is carried on the attendee's ticket. The event organizer can click a sold seat to view the attendee and order details, can create a manual order for chosen seats, and can transfer an attendee to a different seat.	Seat allocation is either self-selected by the attendee at purchase or assigned manually by the organizer, and no automatic or rule-based allocation driven by participant attributes is documented. There is no configurable constraint layer, no measurable optimization of the resulting arrangement, and no repair workflow that preserves the remainder of a published plan when one attendee changes.

2.4 White Space Analysis

Table 2.2 below analyses the reviewed existing systems solution in the market against the five (5) capabilities that the PJKIT use case requires as below:

- i. CP1: Participant-facing seat lookup interface
- ii. CP2: Configurable participant profile-based allocation
- iii. CP3: Controlled dynamic reallocation upon ad hoc event
- iv. CP4: Auto generation of seating plan with constraint-based optimization
- v. CP5: Supports PJKIT event specific structural rules for seating allocation

Every assessment is based on the publicly available product documentation, feature pages and help articles of each vendor, inspected in June and July 2026. A cell marked "Partial" denotes a capability that exists in some form but is not configurable over domain-specific participant attributes or is not exposed as a controlled workflow. A cell marked "Not documented" denotes a capability that is absent from the reviewed vendor documentation, which is distinguished from a capability verified to be absent.

Table 2.2 Whitespace Analysis on Existing System

System	CP1	CP2	CP3	CP4	CP5
Manual Excel Spreadsheet	No	No	No	No	Yes, only by human effort, and unverifiable
Cvent	Partial, sharable seating plan via email	No	No	No	No
PerfectTablePlan	No	Partial, only guest proximity preferences	No	Yes, genetic algorithm based automatic assignment	No
WeddingWire Seating Chart Tool	No	Partial, only limited set of rules can be customized	No	No	No

Prismm	No	Partial, VIP requirements are treated as annotations	No	No	No
Eventbrite Reserved Seating	Partial, as the seat appears on the attendee's own ticket	No	No	No	No
Proposed System	Yes	Yes	Yes	Yes	Yes

The white space analysis showing above indicates that the reviewed existing systems and software solution excel at visual event planning, guest list and attendance management, manual seat arrangement, drag-and-drop seating and sharable seating map generation. When the Primary goal of this system is to visualize tables, chairs or floor plans, these features are very helpful to the event administrator. However, there are four (4) research gaps that have been discovered based on the white space analysis.

First there is a participant-facing seat accessibility gap on most of the reviewed tools. The existing event seating planners allow seating charts to be printed, exported, shared or linked to tickets, but they didn't include a flow that allows the event participants to self-service on checking the event latest seating plan and their assigned seats details such as seat number before the start of the event. Eventbrite Reserved Seating tools does place the seat label directly on the attendee's ticket, which is the closest equivalent found in the review, but it is built on a model where the system allow event participants to select their own seat at the point of purchase rather than the desired model where all the seats to event participants are allocated by the event administrator then published an approved version of the seating plan.

Secondly, there is a controlled dynamic reallocation gap. Dynamic changes such as participant absence or a last-minute substitution are typically handled by manual editing rather than by a systematic reallocation method to the latest published seating plan. Eventbrite Reserved Seating allows an event organizer to transfer an attendee to a different seat, Cvent and PerfectTablePlan allows only the last minutes

editing with synchronization, but neither of the reviewed solutions treats the latest published plan as a baseline to be preserved. Configurable constraint weights also does not support well in the existing systems, which make the event administrator unable to formally adjust the relative importance of competing allocation preferences to generate a desired seating plan.

Next, there is a configurable participant seating allocation gap across all the reviewed tools, which means that most tools do not support automated seating allocation and generation of seating plan that is driven by comprehensive participant profiles and other key qualities such as contribution tier, priority, activeness and event-specific restrictions. Also, none of the reviewed system clearly provide a generalized rule layer that allows the event administrator to configure the hard constraints and weighted soft constraints over domain-specific attributes such as the contribution amount, and activeness score. PerfectTablePlan only allow the configuration for the participant proximity for the seating plan generation.

Lastly, none of the reviewed systems aside from manual excel spreadsheet workflow can express the PJKIT event-specific structural rules and include them as the criteria for the generation of seating plan. This indicates that none of the reviewed system able be extended from their current generic event management and seat planning capabilities, which means that the existing solutions could not be customized for seating plan generation of the PJKIT business use case.

2.5 Comparison of Seating Allocation Optimization Methods

Based on the research study, multiple algorithmic approaches could be implemented for the creation of the seating allocation solver function. Table below compares the reviewed seating allocation method from previous studies.

Table 2.3 Seating Allocation Optimization Methods

Method	How It Works	Strength	Limitation
Greedy priority based	Participants are sorted by priority and assigned the best available seat one by one.	More rule-aware than random or first come-first-served.	Early locally good decisions can force poor later assignments; weak with competing constraints.
Genetic Algorithm (GA)	Evolves a population of seating plans via selection, crossover, and mutation	Flexible for large search spaces and complex scoring; used in seating tools such as PerfectTablePlan (Oryx Digital Ltd., 2026).	Typically, cannot prove mathematical optimality; requires tuning; no falsifiable status reporting.
Integer Linear Programming / Mixed-Integer Programming (ILP/MIP)	Represents the problem with linear equations, integer variables, and objective function.	Mature exact optimization with strong solvers when the model is linear. It is used as one of the algorithms for resolving the seating sub-problems of Ipsen et al. (2026).	Logical and conditional rules such as pairing, adjacency, if-then eligibility, and aisle exclusion must be manually compiled into linear form, typically via the big-M constructions that are error-prone and weaken relaxations.
Constraint Programming – Satisfiability (CP-SAT)	Represents the problem with integer variables, native logical constraints, Boolean indicators, and an objective function. It solves via lazy clause generation (Stuckey, 2010; Perron et al., 2023).	Native reification and channeling for if-then rules. No big-M compilation and have proven optimality with explicit status reporting including OPTIMAL, FEASIBLE, INFEASIBLE, MODEL_INVALID, or UNKNOWN.	Requires careful integer-based modelling; optimality is claimed only when status is OPTIMAL

From the Table 2.3 above, the Constraint Programming – Satisfiability (CP-SAT) is selected as the seating allocation optimization method that to be implemented in this project as the core of the solver function. Further elaboration is mentioned in Section 3.3 Methods and Technologies Adopted.

2.6 Literature Synthesis

By synthesizing the reviewed academic studies and the reviewed existing systems against the three (3) established problem statement define in Section 1.2, it concluded that none of the reviewed academic studies or existing market systems could resolves all the three PJKIT business use case problem statements identified in this project.

In conclusion, the research gap is a combination gap rather than a single missing feature on the existing systems. The individual capabilities exist separately across the market, PerfectTablePlan performs automated generation of seating plan, Eventbrite Reserved Seating Tools gives participant the number and details of their seats, and WeddingWire Seating Chart Tools allow the seating plan to be sharable to the participants via email. However, none of the reviewed academic studies or existing market system provides a solution that exactly suit for the PJKIT business use case. Hence, this final year project is aimed to address this identified combination gap by design and developing an intelligent seating allocation system that take in account on the event participant profile data, predefined hard constraint, configurable weighted soft-constraint to generate an optimized seating plan for PJKIT’s annual events, while providing movement-minimizing repair of an already-published seating map upon ad hoc participant data changes and closes the operational loop by ensuring that participants could retrieves and view the latest seating information.

CHAPTER 3

METHODOLOGY

3.1 Software Development Methodology: Agile SCRUM

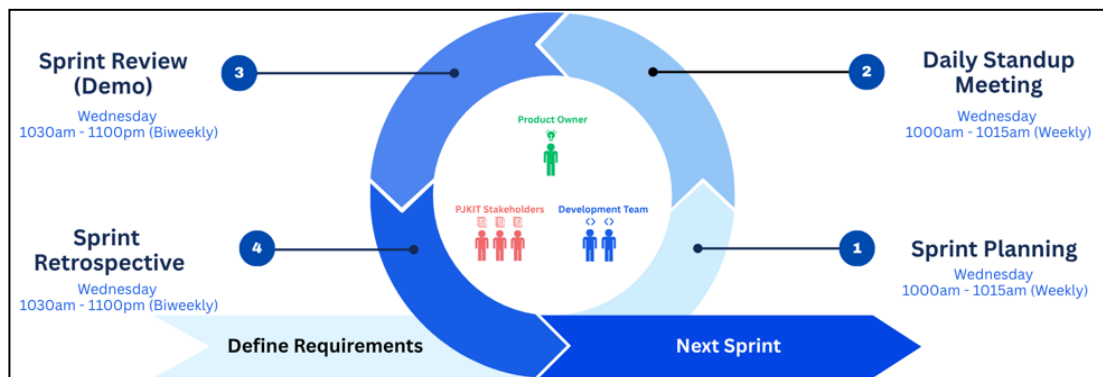


Figure 3.1 Agile SCRUM two-week sprint with NX industry supervisor

This final year's project implemented the Agile software development methodology, specifically the SCRUM framework (Schwaber & Sutherland, 2020) together with the Netizen eXperience (NX) assigned industry supervisor, Mr. Cheah Poh Xiang. Based on the Figure 3.1 above, each sprint in Agile SCRUM framework consists of four SCRUM ceremonies as defined in the official SCRUM Guide (Schwaber & Sutherland, 2020), which including the Sprint Planning for backlog brainstorming and task effort estimation, followed by Daily Standup Meeting for ongoing progress synchronization, and closed by a Sprint Review to demonstrate the developed increment of the intelligent seating allocation system, and Sprint Retrospective that discover the possible improvement steps derived from goods and bad practices observed during a

sprint. All the ceremonies in Final Year Project 1 (FYP1) phase are held on Wednesday with the industry supervisor and the developer that are currently maintaining the Netizen eXperience Event Management platform for PJKIT. The project is implementing a two-week sprint setting, which Sprint Planning and the Stand-up meeting run weekly from 10:00am to 10:15am to set and update the sprint backlog status, while the more important Sprint Review and Retrospective run biweekly from 10:30am to 11:00am to demonstrate the increment developed in the sprint and reflect on the sprint.

3.2 Software Design and Architecture

3.2.1 System Use Case Diagram

The Figure 3.2 shows the system use case diagram for the proposed intelligent seating allocation system that having four (4) actors. The two (2) primary actor that being the main user that directly interact with the system are the Event Administrator and the Event Participant. Event Administrator can view participant list retrieved from the secondary actor, which the Netizen eXperience Management Web App, then select required participants for generating the seating plan with another secondary actor of the system which the deployed Seat Allocation Solver Function. Event Administrator could optionally configure the constraint weightage for the seating allocation system before the generation of new seating plan. Upon the generation of the seating plan, Event Administrator can review the version and decided whether to publish the seating plan to the participant flow. If there are the occurrence of the ad hoc event that causing the participant data to be changed, Event Administrator could regenerate ad hoc seating plan by including the latest seating plan as the baseline for repair. Another primary actor of the system is the Event Participants, which they can view their assigned seats detail and view the latest published seating map in the system with the precondition of a published plan has existed in the system.

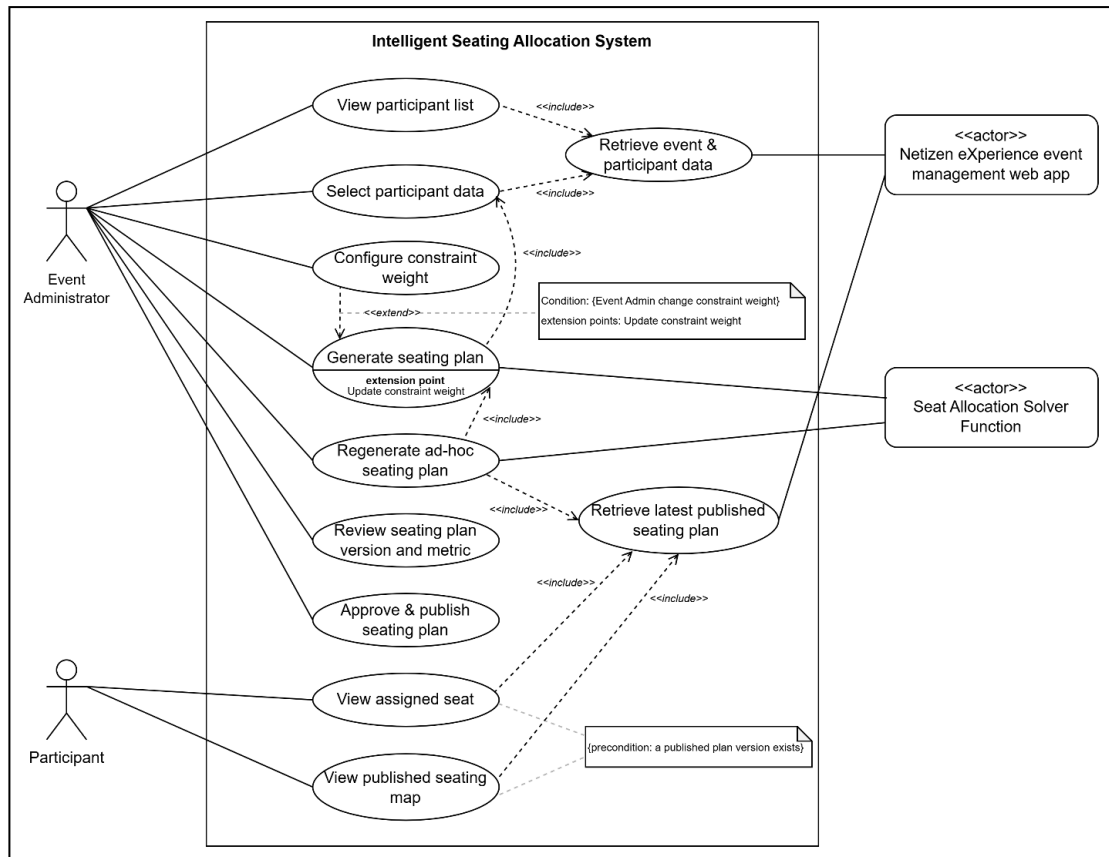


Figure 3.2 System Use Case Diagram of Intelligent Seating Allocation System

Table 3.1 Definition for Generating Seating Plan Use Case

Field	Description
Use Case	Generate Seating Plan
Primary Actor	Event Administrator
Supporting Actor	Seat Allocation Solver Function
Goal Trace	G1 (G1.1–G1.3)
Precondition	The system should receive selected participant data, venue seating layout, and constraint configuration.
Trigger	Organizer clicks <i>Generate Seating Plan</i> in the admin module.

Main Success Scenario	1. Platform packages participants, layout, configuration, and optional previous allocation into a JSON payload. 2. Engine validates schema, derives the demand-driven tier bands, and checks band capacity. 3. Engine builds the integer constraint model over eligible combinations only. 4. CP-SAT solver function solves the seating allocation within the configured time budget. 5. Engine extracts assignments and penalty breakdown. 6. Independent validator rechecks all the hard constraints and the reconstructed objective. 7. Result is stored as a new seating plan version. 8. Organizer reviews the version and quality indicators.
Extensions	2a. Invalid payload return structured INVALID_INPUT error. 2b. Capacity exceeded return a structured infeasibility explanation message without invoking the solver. 4a. INFEASIBLE / UNKNOWN status and conflict hints returned, the organizer relaxes rules or adjusts seat range. 4b. FEASIBLE with proven optimum required reported as OPTIMAL_NOT_PROVEN 6a. Independent validation fails lead to result rejected, failure reported.
Postcondition	A reviewable plan version exists; nothing is published without explicit approval (UC6).

3.2.2 System Flowchart

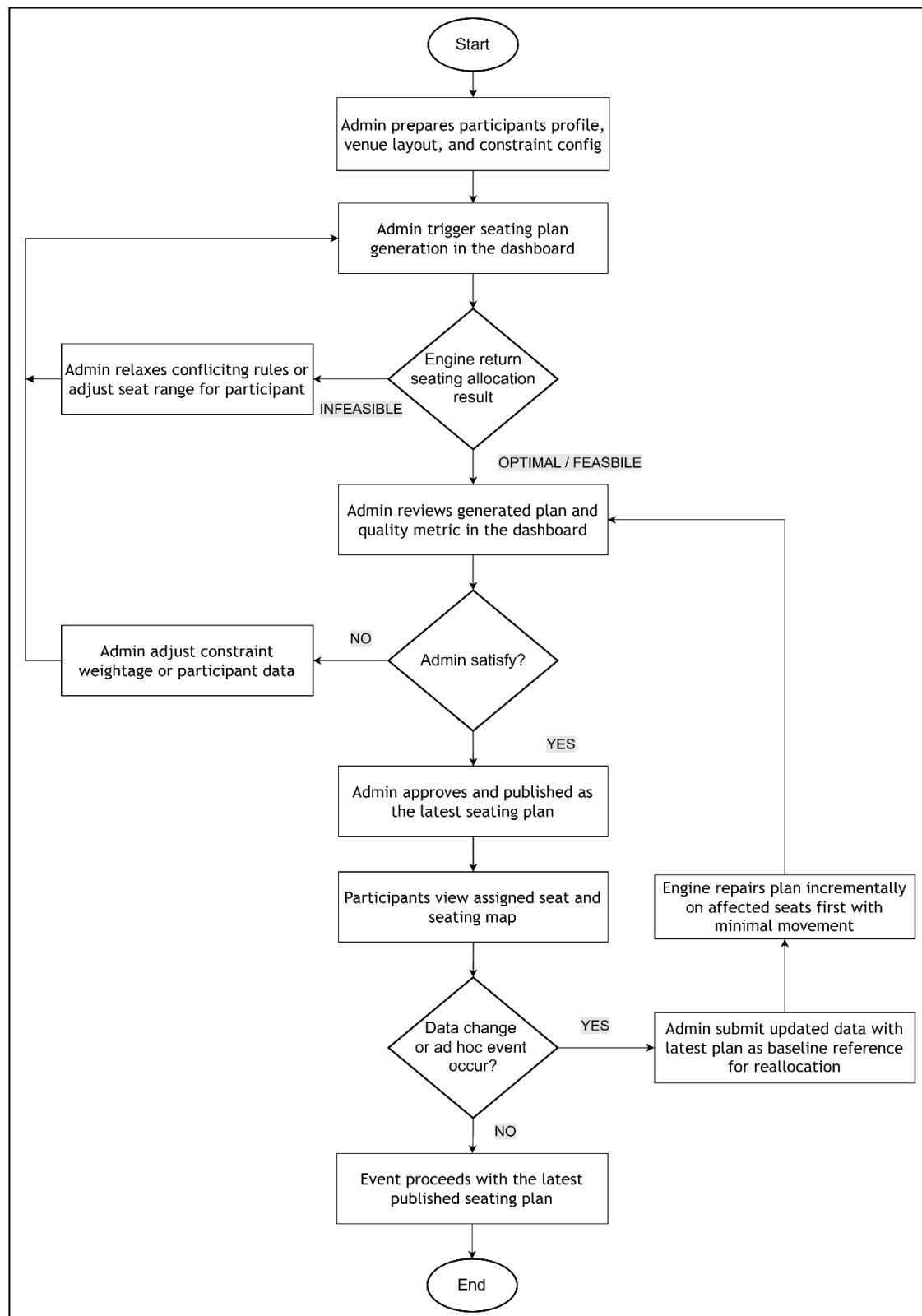


Figure 3.3 System Flowchart of Intelligent Seating Allocation System

3.2.3 System State Machine Diagram

Based on the system state machine diagram shows in Figure 3.4, the system state diagram demonstrated the life of one seating allocation session in the seat allocation solver function. First, the serverless aspect of the engine is illustrated in the entry transitions, which a first invocation of the solver function requires to transition into a ColdStart state before starting to solve the seating allocation.

The Solving State fork into two (2) different states, either ResultReady state when the solver returns the status of OPTIMAL or FEASIBLE and generates a seating map solution, or the Failed state when the returned solver status is INFEASIBLE, INVALID or TIMEOUT. Also, moving the ResultReady to the UnderReview state, the system state can branch into two (2) different state, which either back to the Solving state when the event administrator choose to regenerate the seating plan again, or forward to the Published state when the event administrator approved the result and published the seating map to participant.

The most important part to be observed from the state machine diagram above is the seating map reallocation looping, where a Published seating plan can be Reloaded and fed back into the Solving state as reference, so that the incremental repair of the seating plan could minimize the movement of the participant. The session thus only will be terminated from the Published state, when the event administrator confirmed that the latest published seating plan is in the single stable resting state of the system.

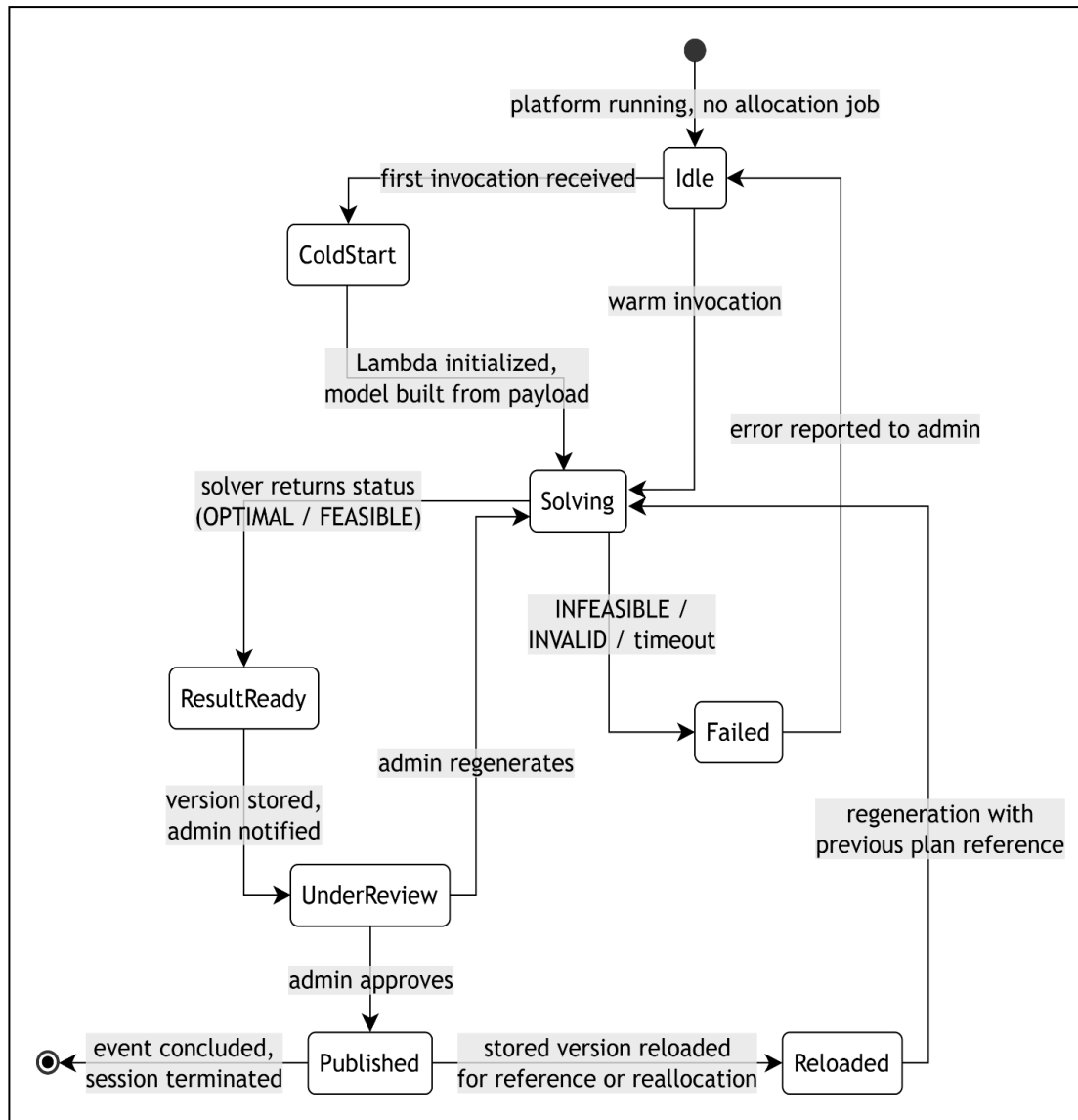


Figure 3.4 System State Machine Diagram for Seating Allocation Solver Function

3.2.4 System Architecture Diagram

The proposed intelligent seating allocation system in this project adopts a four-layer architecture model that consists of the layered architecture pattern commonly used to separate presentation, application, compute, and data concerns in distributed systems. The architecture model separates the actors who use the system from the internal or external software that they interact with, a computation layer that focuses on

generating the seating plan at serverless compute side, and the storage that persists the required metadata for event, participant, and seating allocation result.

Based on Figure 3.5 which shows the layered model system architecture diagram, the first layer of the architecture model is the Users layer that holds the two primary actors of the system, which are the Event Administrator and the Event Participant. The second layer is Application Layer, which is the Netizen eXperience Event Management Platform, containing the Admin Seating Allocation Dashboard, a Next.js server component that will call the server actions that will interact with the deployed AWS resources via AWS SDK. The Application Layer also contains a Participant Seat Lookup Interface that will fetch and display the latest participant seating details on the latest published seating map retrieved from DynamoDB. The Compute Layer is focused on the AWS serverless, where an AWS Lambda function hosts the core seating allocation engine and its embedded Google OR-Tools CP-SAT solver function (Amazon Web Services, Inc., 2025; Google, 2024). The Data Layer is AWS storage, comprising Amazon DynamoDB to store event, participant, and allocation metadata, along with the latest-approved seat references. Amazon S3 is also being deployed to Data Layer to store the versioned seating-plan result JSON for seating plan generation record traceability and referencing the latest seating plan. The implementation of the layered architecture model ensures that each tier depends only on the one above it, so the application, the solver function, and the storage can be scaled and modified independently.

The primary flow begins with the Event Administrator. From the Admin Seating Allocation Dashboard, the organizer optionally configures the constraints and selects the range of participant data to be involved in the event seating allocation. Then, the Next.js server component transforms this data into a structure JSON file that will be used to invoke the Lambda function in the Compute Layer through the AWS SDK (Amazon Web Services, Inc., 2015). The seating allocation engine builds and solves the model with CP-SAT and returns the optimal/feasible solution together with its solver status inside the Lambda function. The resulting seating-plan JSON will then send back as response to the server component that invoke the solver function, which

triggers server actions to store the event and participant metadata in DynamoDB and the versioned seating-plan document in S3.

When a change occurs after publication, the same path is used for reallocation where the server component re-invokes the Lambda function with the updated payload and the previous seating plan, so that the engine will perform repairs and minor adjustment on existing seating plan rather than regenerates the whole seating plan with all the same data payload. The Event Participant never reaches the compute tier, the Participant Seat Lookup Interface requests participant-specific seating details from the server component, which serves the latest published seating details read from DynamoDB, and the participant could view their seat. The takeaway is that computation is triggered only by the administrator and runs on demand, while the participant's path is a short, read-only lookup of already-approved data.

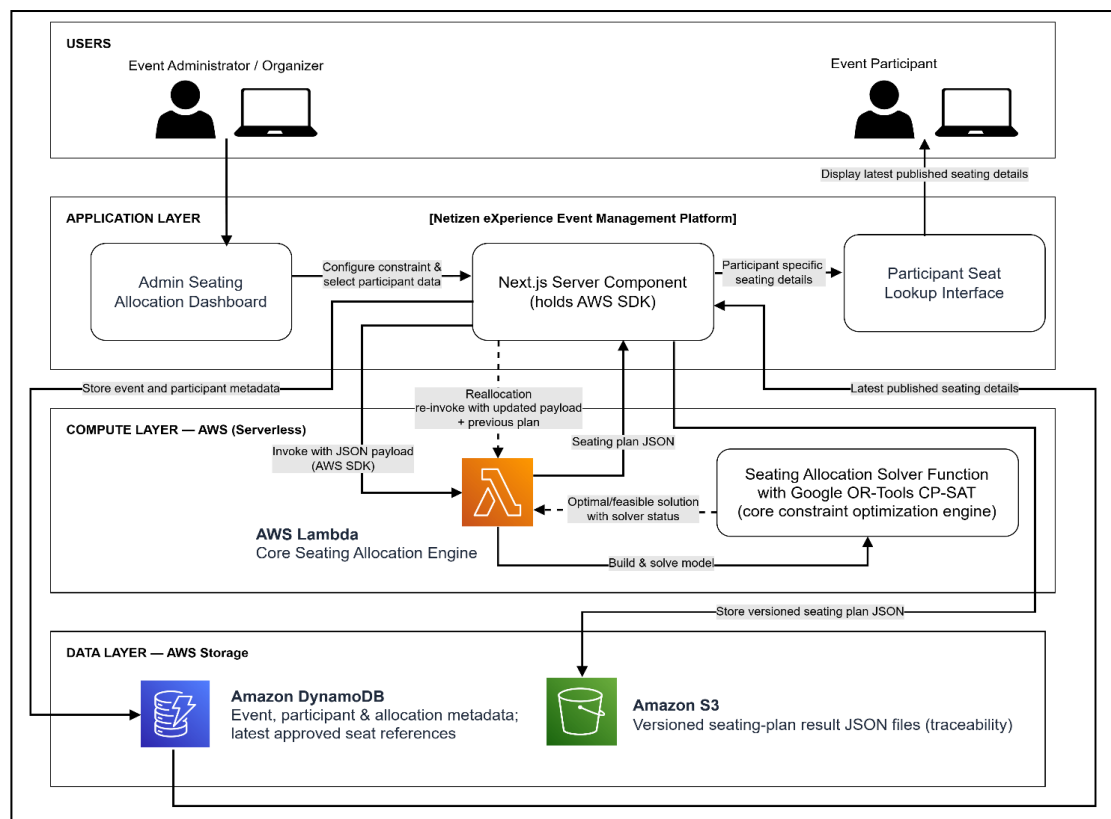


Figure 3.5 System Architecture Diagram of Intelligent Seating Allocation System

3.2.5 Class Diagram

Based on the Figure 3.6 that showing the Class Diagram of the proposed Intelligent Seating Allocation System, the data modelling structure started from the Event class as the root, which an event may be registered by many Participants, and each event will only use one single venue layout. Every event is configured by one ConstraintConfig class, which consists of HardConstraint and SoftConstraint definition. Each event may produce many AllocationPlanVersion that represent the seat assignment results from the generation operation.

There is also a self-reference relationship on the AllocationPlanVersion class, which lets the reallocation compare and measure the movement against the prior published seating plan. Each Seat class carries the physicalPosition and priorityRank as separated attributes to separate the aisle logic from the preference logic. The solver function engine-side SolverEngine and ConstraintModelBuilder are cleanly separated from the domain and artifact classes by implementing the Builder design pattern, so that the model's builder operations are concrete realization of the constraint model, but not a property of the domain data.

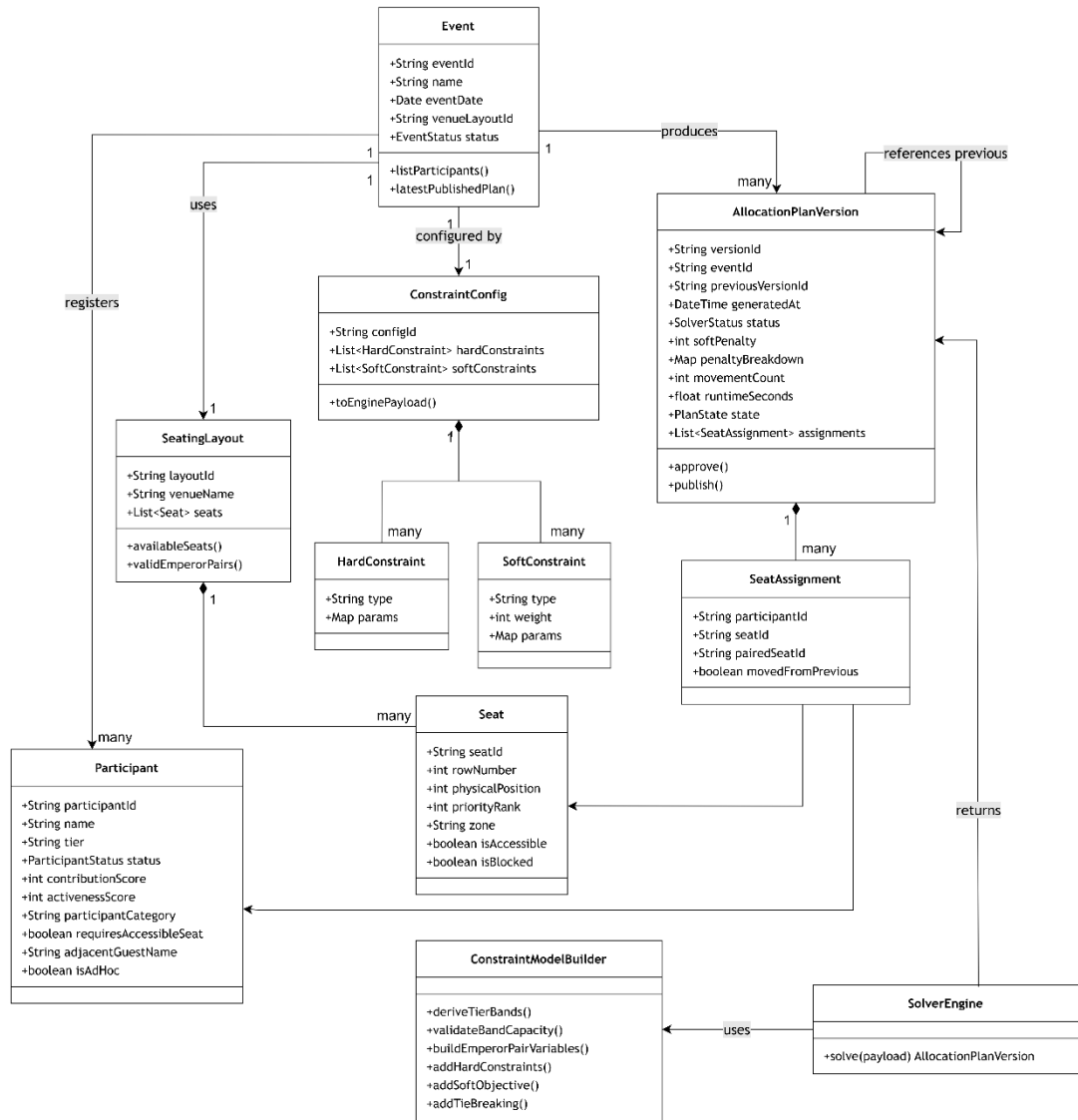


Figure 3.6 Class Diagram of the Intelligent Seating Allocation System

3.2.6 Swimlane Diagram for Generate Seating Plan

Based on the activity diagram showing in the Figure 3.7, it involved four (4) swim lanes which the event administrator as actor to initiate the flow of generating seating plan, Netizen eXperience Event Management Web Application as the second swim lane that focus on the forwarding the incoming request of generating seating plan from Event Administrator to Intelligent Seating Allocation System, and returning any seating plan generation result or errors from Intelligent Seating Allocation System

back to the event administrator seating allocation operation dashboard interface. The third swim lane which is the Intelligent Seating Allocation System act as layer to validate the incoming input data such as participant profiles, seating layout and constraint weights and the main system to invoke the deployed Seating Allocation Function which the fourth swim lane to get the result of the seating map generation.

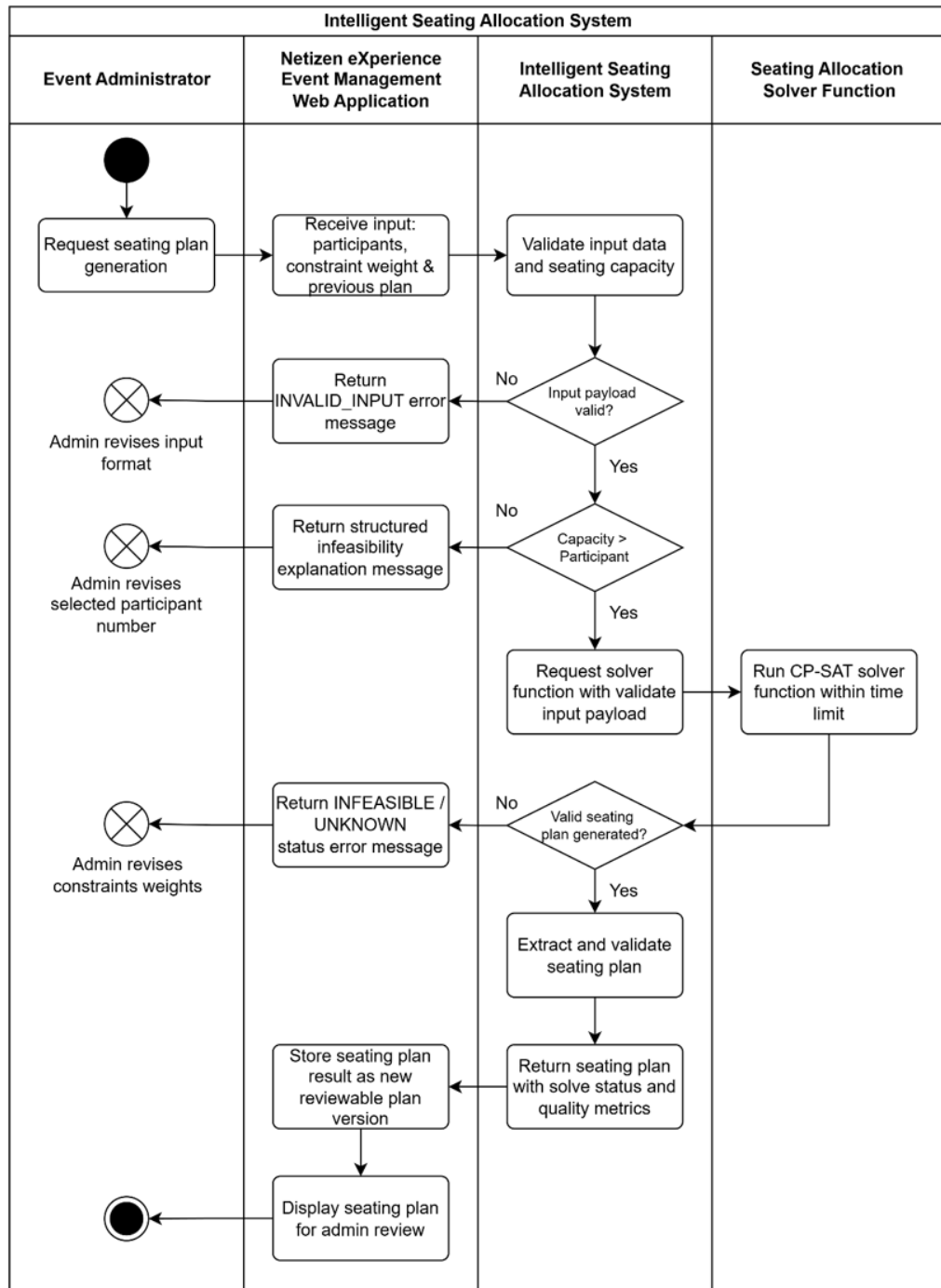


Figure 3.7 Activity diagram of the generate seating plan use case

3.3 Methods and Technologies Adopted

Table 3.2 Methods and Technologies Adopted

Method / Technology	Role in the Project
Constraint Programming (CP) with hard and weighed soft constraints	Core method used to model event seating rules as a Constraint Optimization Problem.
Google OR-Tools CP-SAT solver	The library used to execute the constraint model and report explicit solver statuses (optimal, feasible, infeasible, model invalid, unknown).
Python 3	Programming language used for the implementation of the seat allocation solver function.
AWS Lambda	An on demand serverless deployment of the engine, invoked with an explicit solver time budget.
AWS SDK (from the Next.js service component)	Invocation channel for AWS infrastructure with JSON payloads and results.
Next.js / React (TypeScript)	Existing Netizen eXperience Event Management platform that is currently hosting the admin module and integration with a participant lookup route as the outcome of the project.
Amazon DynamoDB & Amazon S3	Persistence of allocation metadata and versioned plan JSON respectively.
SST Framework	Infrastructure-as-code framework used to provision the AWS resources.
GitHub Issues & Projects	Tracking all the backlog and task for the development of intelligent allocation system in the Netizen eXperience repository.

Based on the Table 3.2 above, Google OR-Tools CP-SAT solver is selected as the solving engine in this project for the seating allocation problem is because it accepts integer decision variables, Boolean indicators, and conditional constraints through reification and channeling, and reports explicit solver statuses that this project surfaces as part of allocation quality (Google, 2024). This selection is further grounded in the

primary technical literature describing CP-SAT's lazy-clause-generation architecture, which combines SAT-style conflict-driven clause learning (Stuckey, 2010) with linear-programming relaxation bounds and portfolio parallel search (Perron et al., 2023) an architecture specifically designed to prune search spaces of this project's scale while still proving optimality, which is a behavioral that a guarantee custom greedy or metaheuristic approaches cannot offer.

3.4 Seating Allocation Optimization Lifecycle

The seating allocation optimization lifecycle describes how a plain human language event seating rule from PJKIT event administrators transform into a validated seating map as the production result. It is also referred to the process that the solver function implements. The lifecycle has designed into seven (7) stages, where each stage is a distinct transformation with its own output, and the output of the current stage will be the input to be used in the next stage.

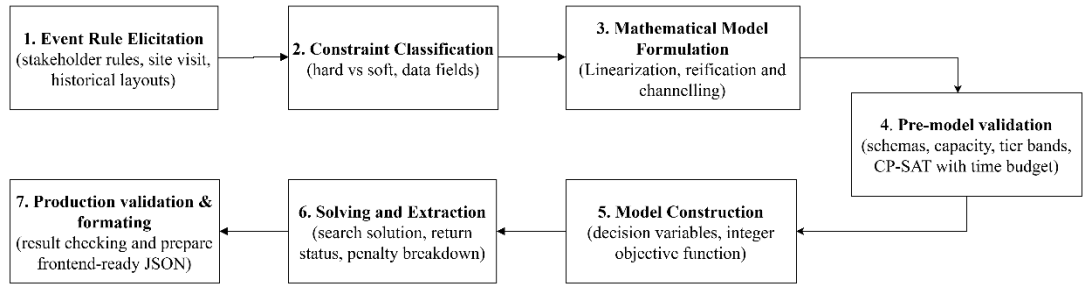


Figure 3.8 Seating Allocation Optimization Cycle

The first stage of the optimization cycle is event rules elicitation, where the event specific seating rules are gathered from the PJKIT stakeholders directly through the site visit, interview, the joint application development workshop and the direct observation on the creation of event seating plan as described in Section 3.5.1.

The second stage is constraint classification, where each of the elicited rule listed in the previous stage will be classified either as a hard constraint that every valid seating map must satisfy, or as a weighted soft constraint that represents a preference of the event administrator that could be violated based on a specific amount of cost value as penalty. The output of this stage is the constraint listing in Section 3.5.3 which the Table 3.4.

The third stage is mathematical formulation, where each classified constraints from the event rules is further expressed as a mathematical statement over decision variables by using the method such as linearization, reification and channelling principles. The output is the constraint optimization problem (COP) formulation and mathematical expressions to be used for model construction.

The fourth stage is pre-model validation, where the incoming payload from the system is checked against its predefined schema for the purpose of type checking, the contribution tier band are derived from the registration demand of the event, and the capacity of each band is verified before any model is constructed for the seating map generation. According to the elicited functional requirements FR15 listed in the Appendix A, a failure at this stage returns a structured error to the admin without invoking the solver function.

The fifth stage is model construction, where only the eligible participant-seat combinations allowed by the defined participant contribution tier band are used to produce the decision variables. Then, the deterministic tie-breaking technique is combined with the weighted soft constraint penalties and the hard constraints to create an integer objective function that used to evaluate the optimality of the result.

The sixth stage is solving and extraction, where the CP-SAT solver function searches the feasible and optimal seating plan within the configured time budget and returns both the seating plan solution and an explicit solver status. In order to ensure that a FEASIBLE state seating plan result is never returned as an optimal solution by the solver function, both the assignments and the per-constraint penalty breakdown are retrieved from the solution while maintaining the status.

The seventh stage is the production validation and formatting, where the extracted seating map is re-checked against every hard constraint by a separated validator implemented independently of the solver, including a reconstruction of the reported objective value. After that, the result is then transformed into the frontend-consumable JSON structure that could be used to render the seating plan and stored as a new plan version in the system.

In conclusion, the Seating Allocation Optimization Lifecycle is designed as an iterative procedure where if the PJKIT stakeholders reveals that a modelled constraint behaves differently from the community's intent and their desired outcome, the affected constraint will be modified based on the feedback and then returns to the second and third stages for reclassification or reformulation, which is the mechanism by which the constraint listing is refined across sprints.

3.5 Preliminary Works

3.5.1 Interview and Joint Application Development Workshop



Figure 3.9 Site visit to PJKIT for stakeholders' meetup

A site visit to the PJKIT was conducted together with the industry supervisor from Netizen eXperience on 26th June 2026. The site visit combined a semi-structured interview with PJKIT stakeholders with the elicitation questionnaires in Appendix A, and direct observation of the event venue physically.

At the same day, a joint application development (JAD) workshop is carried out between the FYP student, Industry Supervisor – Mr. Cheah Poh Xiang and three (3) representatives including the head of administrator that handle on event seating allocation from PJKIT side to brainstorm, discuss and refine the requirement for the proposed intelligent seating allocation system. The result of this elicitation activity is located at Appendix B.

3.5.2 Preliminary Findings from Elicitation

The validated constraint and scenario base that underpin the entire requirement set were obtained via the elicitation activities in Section 3.5.1.

Table 3.3 Preliminary Findings from Elicitation Activities

Preliminary Finding	Description
1. Validated event participant scale and venue layout	PJKIT's event typically includes 100 – 200 participant per event, and the participants is allocated in a 16 rows × 16 seats venue layout which in total of 256 numbered seats, where 232 assignable after 24-seat is blocked by the building structure in the center of the hall. Participants are arranged in a concert-style with a numbered front section and a free-seating section at the back of the venue.
2. Existing workflow for generating seating plan is conducted manually by event	PJKIT currently prepares all the seating maps for every event manually by inserting participant name one-by-one into the predefined seating chart in an excel sheet. The seating map generation procedure is carried out by a

administrator and volunteers.	team of fewer than five senior volunteers who must personally know the participants and understand the significance of each placement. The current workflow takes up until two (2) weeks before the event day.
3. Contribution tier structure as the first-order rule for seating allocation	Participants are first categorized by their contribution tier which either Emperor, Merit, or Bodhi tier. The seating allocation ordered as the last row of Emperor tier should precede the first row of the Merit tier, and the last row of Merit tier should precede the first row of Bodhi tier. The seating row could be shared between two different tiers to avoid empty seats in the numbered front row.
4. Generic tools in the market cannot be used to solve the specific seating allocation problem in PJKIT	The specific rules of PJKIT seating allocation for participants increased the necessity to implement a customized solution that could generate a seating map by satisfying all the structural rules such as contribution tiering priority, side-of-hall placement of participants who need accessible seats and within-tier contribution ordering.
5. Dynamic participant data change on ad hoc event for PJKIT	PJKIT will close the registration of the event on a set date, walk-in and late registration are uncommon scenario to occur in PJKIT event. However, the primary ad hoc event case for PJKIT is the absence of a registered participant, sometimes also with replacement participant to take the vacated seat on the event day. In every case the stakeholder expectation is that a published plan is repaired with minimal disruption instead of regenerating the whole seating map again.

3.5.3 Event Rule Listing and Constraint Classification

The Table 3.4 below shows first draft of the list of event rules collected from the elicitation activities with the stakeholders and industry supervisor in during the site visit to PJKIT. Each of the event rule is further classified into hard constraints or soft constraints as the output of the constraint classification stage.

Table 3.4 Event Rule Listing and Constraint Classification

Ref	Event Rule from PJKIT	Classification
C1	Each participant is assigned exactly one allocation unit (one seat; one valid seat pair for Emperor registrations).	Hard
C2	Each seat is occupied by at most one participant (both seats of an assigned Emperor pair count as occupied).	Hard
C3	Blocked or unavailable seats must not be assigned.	Hard
C4	Assignments must fall within the valid seat range of the event layout.	Hard
C5	Only participants of eligible status may be seated.	Hard
C6	Participants with higher contribution/priority scores should receive seats of higher priority rank.	Soft (weighted)
C7	Participants with higher activeness scores should be favored within their tier.	Soft (weighted)
C8	Participants should be seated in zones suitable for their category (within-tier suitability preference).	Soft (weighted)
C9	Upon reallocation, unaffected participants retain their previous seats by default; unavoidable movement is minimized in number and then in	Soft (weighted)

	distance (movement minimization; penalty is zero when no previous allocation exists).	
C10	An Emperor registration occupies exactly two adjacent seats forming one of the venue's valid within-row pairs; when no adjacent guest name is provided, both seats display the primary participant's name.	Hard
C11	No Emperor pair may straddle the center aisle: positions 8 and 9 never form a pair. Physical adjacency is determined by physical seat position, never by priority rank.	Hard
C12	Tier-band placement: tiers occupy contiguous, demand-derived row bands ordered Emperor → Merit → Bodhi from the front of the hall; a boundary row may be shared between two adjacent tiers (when one tier ends partway along a row, the next continues in it) so the numbered section packs with no empty seat; band capacity (with Emperor units counting two seats) is validated before model construction.	Hard
C13	Within each tier, a participant with a higher contribution score is never seated in a later row than a participant with a lower score.	Hard
C14	Among plans of equal weighted penalty, a deterministic tie-breaking rule selects a canonical plan; the tie-break term is scaled so it can never override any weighted penalty.	Modelling rule (objective)
C15	Front-to-back packing where the numbered section fills seating from front to back with no empty seat (the remaining assignable rows form the free-seating section).	Hard
C16	Centre-out fill where each side of a row fills outward from the center aisle, so no gap appears between the aisle and an outer occupied seat.	Hard

C17	A participant who requires an accessible seat is placed at a side/edge seat of the hall (for an Emperor unit, a pair at the side), for ease of access.	Hard
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3.6 Project Timeline

3.6.1 Final Year Project 1 Gantt Chart

Based on the Figure 3.10, it shows the overall timeline and milestones for the Final Year Project (FYP) phase 1, which from the beginning of May 2026 to the end of July 2026. The development milestones for FYP1 are as below:

- i. Finalize the title and scope of the project.
- ii. Complete literature review on existing studies or solutions in the market.
- iii. Complete the requirement elicitation activities.
- iv. Complete initial event rule listing and constraint classification.
- v. Complete preliminary mathematical formulation for Constraint Optimization Problem (COP) based on the identified hard and soft constraint.

No	Activities	Week													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Project Title Confirmation and Form 01A Submission	■	■												
2	Project Discussion with Supervisor		■	■	■										
3	Preliminary Research		■	■	■	■	■	■							
4	Research on Methodology and Technology Stack				■	■	■	■	■						
5	Progress Update with Supervisor					■									
6	Proposal Defense						■	■							
7	Expectation Analysis & Constraint Gathering						■	■	■						
8	System Design: Architecture, Data Model and Workflow						■	■	■	■	■				
9	FYP Interim Report Preparation					■	■	■	■	■	■	■			
10	Submission of Draft Interim Report											■	■		
11	Sprint 1: Initial Constraint Listing and Classification											■	■	■	
12	Submission of Interim Report and Progress Assessment 2												■	■	
13	Sprint 2: Constraint Optimization Problem Modelling and Boolean Formulation													■	■
14	Biweekly Review with FYP Supervisor	■	■	■	■	■	■	■	■	■	■	■	■	■	■

Figure 3.10 Final Year Project Phase 1 Gantt Chart

3.6.2 Final Year Project 2 Gantt Chart

Figure 3.11 shows the overall timeline and milestones for the next phase of the Final Year Project (FYP), which is phase 2 that starting from beginning of September 2026 to the end of November 2026. The development milestones for FYP2 are expected as below:

- i. Develop and deploy the CP-SAT Solver Function to AWS Lambda.
- ii. Design and develop the seating allocation operation dashboard.
- iii. Design and develop the participant seat lookup interface.
- iv. Develop the dynamic reallocation of seating map on ad hoc event.

No	Activities	Week													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Sprint 3: Development of CP-SAT Solver Function														
2	Sprint 4: Testing and deployment of the Seating Allocation Solver function to AWS Lambda														
3	Sprint 4: Dynamic Reallocation on ad hoc event module														
4	Sprint 5: Web Platform Integration: Admin Dashboard														
5	Sprint 5: Web Platform Integration: Participant Seat Lookup Interface														
6	Sprint 6: System Testing and Evaluation														
7	FYP 2 Dissertation and Technical Paper Preparation														
8	FYP 2 Dissertation and Technical Paper Review with FYP Supervisor														
9	Submission of Dissertation (Softcopy) and Technical Paper														
10	FYP VIVA Preparation														
11	FYP VIVA														
12	Project Dissertation Preparation (Hardcopy)														
13	Submission of Project Dissertation (Hardcopy)														
14	Biweekly Review with FYP Supervisor														

Figure 3.11 Final Year Project Phase 2 Gantt Chart

CHAPTER 4

CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This report has presented the planning-phase work of a Final Year Project Phase 1 that addresses a genuine and recurring operational problem which the manual seating allocation for large-scale assembly events. The background established that seating allocation is a multi-criteria combinatorial problem whose solution space grows explosively with participant numbers, and that the case-study organization, PJ Kwan Inn Teng, currently manages events of 100 - 200 participants in a 256-seat venue governed by demand-derived tier bands, two-seat Emperor allocation units, an aisle that breaks adjacency, and contribution-ordering rules — using spreadsheets, printed charts, and PDF listings. Manual allocation is slow and inconsistent between allocators and cannot be objectively justified against the community's own priority rules; late data changes and ad-hoc registrations force disruptive, error-prone rework of nearly finished plans; and participants queue at the venue to retrieve seats from printed lists that may already be outdated. Without the research and development undertaken in this project, these consequences persist and worsen as event attendance grows.

The literature review demonstrated that neither commercial tooling (Cvent Inc., 2026; Oryx Digital Ltd., 2026; The Knot Worldwide Inc., 2026) nor the published seating-allocation literature (Ipsen et al., 2026; Muñoz et al., 2006) offers this combination which is a declaratively configurable, status-reporting, and stability-aware seating allocation engine. The contribution of this project is to develop a configurable, constraint-based seating allocation system whose core is a solver engine

that models real event rules from PJKIT as a Constraint Optimization Problem that can be solved using the Google OR-Tools CP-SAT solver and deployed as a Python script on AWS Lambda function that invoked with JSON payloads directed from the Netizen eXperience event management platform.

The system provides organizer-controlled generation, versioned review and publishing, configurable rules over PJKIT's domain attributes, controlled dynamic reallocation that incrementally repairs the published plan for ad-hoc participants while leaving unaffected participants in their initial assigned seat. This project also aimed to develop participant-facing lookup interface for PJKIT participants to enable them to perform self-checking on the latest published seat from event administrator before or during the event day.

4.2 Future Works and Recommendations

For the future works and recommendations that I will perform in FYP 2 period is as scheduled in the Section 3.6.1 Final Year Project 1 Gantt Chart and the Section 3.6.2 Final Year Project 2 Gantt Chart.

First, I will complete the constraint optimization problem (COP) mathematical formulation for the seating allocation which is proceeding to the Stage 3 of the Seating Allocation Optimization Lifecycle. I will be completing the mathematical model that include the representation of hard constraints and the objective function. Also, I will extend this preliminary formulation further to cover the formalizing the demand-driven contribution tier grouping, the front-to-back packing and the centre-out fill rules, the accessibility placement at the side of the venue, and the movement minimizing objective function used during the dynamic reallocation. I will document the finished model as a programmable model specification so that every constraint classified in the Section 3.5.3 Table 3.4 can be traced to the mathematical statement that implements it in the solver function.

Moreover, after the completion of the mathematical formulation for all the constraints and objective functions for the COP, I will start the development of the solver function by implementing the modelled hard constraints and the weighted soft constraints using Python with the Google OR Tools CP-SAT library. After implementing all the modelled constraints into the solver function code, I will create a test case scenario with an average of 150 participants who have to be assigned within the event space that have 232 assignable seats to evaluate the functionality of the solver function, which is expected to return an OPTIMAL status seating plan within the configured solver budget. After that I will deploy the solver function to AWS Lambda and measure the cold-start behavior, because the OR-Tools library adds a large deployment artifact and the on-event-day repair case is the most time-sensitive use case of the system upon the trigger of the reallocation.

I will also keep participating in the Agile SCRUM ceremonies with the industry supervisor, Mr. Cheah Poh Xiang following the designed two-week sprint arrangement described in Section 3.1. By actively joining the ceremonies, I could ensure that all the increment of the development such as the mathematical formulation and solver function will be demonstrated, reviewed and getting valuable feedback from Mr. Cheah Poh Xiang for technical advice and reconciliation on the requirement from PJKIT side. On the other hand, I will begin reviewing the current codebase of the Netizen eXperience event management web application, which mainly focuses on Next.js framework, the existing participant and event data model, and the AWS infrastructure and stacks provisioned for the web application. The purpose of the codebase review is to identify the integration points for the administrator seating allocation dashboard and the participant seat lookup rote before the integration phase begins in the second half of FYP 2 period, so that the integration is not delayed by unfamiliarity with the existing platform.

To conclude, I will be following the proposed project timeline in Section 3.6 for future development task in FYP2 period and continue refining the requirements and the constraints with PJKIT. I will be conducting monthly site visit to PJKIT together with Mr. Cheah Poh Xiang to showcase the current progress of the seating allocation system and to validate the modelled rules and constraints.

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APPENDICES

APPENDIX A: Consolidated Functional Requirements

ID	Functional Requirement Description
FR1	The system shall accept participant profile data from the existing web platform as a structured JSON payload.
FR2	The system shall accept seating layout data from the existing web platform as a structured JSON payload.
FR3	The system shall allow event organizers to configure the weights of the supported soft constraints.
FR4	The system shall map the configurable event rules into computable hard constraints and weighted soft constraints.
FR5	The system shall generate a seating allocation result based on participant profiles, seating layout, and constraint configuration.
FR6	The system shall return the generated seating allocation result in a front-end-consumable JSON format, including when the run fails.
FR7	The system shall return the solver status of every allocation run (OPTIMAL, FEASIBLE, INFEASIBLE, UNKNOWN, or error) together with a clear status message when allocation succeeds, fails, or becomes infeasible; a FEASIBLE seating allocation result shall be reported as OPTIMAL_NOT_PROVEN when a proven optimum is required.
FR8	The system shall calculate and return reproducible quality indicators for every generated plan including solver status, independent hard-constraint validation result, total weighted soft-constraint penalty with a per-constraint breakdown, movement count and distance where applicable, and runtime.

FR9	The system shall perform dynamic reallocation as incremental repair of the published plan, modifying only the assignments affected by a change minimizing first the number of unaffected participants moved and shall store the repaired result as a new plan version requiring organizer approval under FR14.
FR10	The system shall allow event organizers to review generated seating plan versions, including the seating map and quality indicators, through the admin review page.
FR11	The system shall allow seating results to be exported for event operation.
FR12	The system shall store and retrieve the persisted entities of the data model: events, participants, layouts, seats, constraint configurations, plan versions, and assignments, with each plan version referencing its predecessor.
FR13	The system shall allow a participant to retrieve their assigned seat and view the seating map from the latest published seating plan.
FR14	The system shall store every generated allocation result as a plan version, shall require explicit organizer approval before publication, and shall expose only the latest published version to participants.
FR15	The system shall validate every incoming payload against its schema and structural rules — including pre-model tier-capacity validation — and shall return a structured, frontend-consumable error response for invalid input.
FR16	The system shall re-check every generated seating plan against all hard constraints independently of the solver that produced it, including reconstruction of the reported objective value, before the plan can be accepted.
FR17	The system shall return a structured failure response containing the solver status and any determinable validation, capacity, or input failure reasons, shall prevent an unsuccessful result from being published, and shall allow the organizer to revise data, constraints, weights, or seat availability.

APPENDIX B: Consolidated Non-Functional Requirements

ID	Non-Functional Requirement Description
NFR1: Performance	The system shall return an OPTIMAL result within the configured 60-second solver budget for the default dataset of 150 participants, which comprises 94 allocation units after the 112 Emperor participants are paired into 56 two-seat registrations, on the 256-seat (232-assignable) case-study layout under the defined evaluation environment; when the budget expires before optimality is proven, the system shall return the actual solver status and shall not represent the solution as optimal.
NFR2: Usability	The system shall enable an event administrator to generate and publish a seating plan without assistance after a single briefing session (criterion subject to stakeholder validation).
NFR3: Reliability	The system shall not publish a plan that fails input validation, solver processing, or independent post-solve validation, and shall keep the previously published plan unchanged and accessible when such a failure occurs.
NFR4: Maintainability	The system shall isolate the constraint configuration from the solver logic such that a change to the constraint configuration requires no modification of solver code.
NFR5: Security & Privacy	The system shall inherit administrative access from the host platform's authentication, shall restrict unpublished plan versions to authorized administrators, shall allow each participant to view only their own assignment from the latest published version, shall use pseudonymous participant identifiers in the solver payload, and shall retain data in line with the event lifecycle.
NFR6: Scalability	The system shall support the validated operational baseline of 150 participants (94 allocation units, 150 occupied seats) on the 232-assignable-seat layout and shall treat larger datasets as exploratory scalability experiments paired with layouts of sufficient capacity.

NFR7: Availability	The participant seat lookup should remain available throughout the event window, the target availability level should be elicited and agreed with Netizen eXperience because the lookup depends on the host platform's availability characteristics.
NFR8: Reproducibility	The system shall produce the same canonical allocation and quality indicators for identical normalized input data, constraint configuration, solver parameters, and deterministic seed, repeated executions.

APPENDIX C System Goal Definitions

Goal ID	Goal Description	Type
MG1	The system shall enable event administrator to produce fair, rule-consistent seating plans for large-scale assemblies (validated baseline: 100–200 participants; 256-seat venue with 232 assignable seats) with reduced manual effort and controlled handling of ad-hoc changes.	Root
G1	The system shall generate constraint-compliant seating plans automatically.	Hard
G1.1	Event rules shall be modelled as computable hard constraints and weighed soft constraints.	Hard
G1.2	Plans shall be generated from participant profiles, seating layout, and constraint configuration.	Hard
G1.3	Every run shall report a verifiable solver status and reproducible quality indicators.	Hard
G2	Published plans shall be reallocated under administrator control when participant or event data changes.	Hard
G2.1	Reallocation shall repair incrementally, minimizing first the number of unaffected participants moved and then total movement distance for the latest published seating map.	Hard

G2.2	Every generated or repaired result shall be a version requiring explicit administrator approval before publication.	Hard
G3	Participants shall retrieve their assigned seat themselves.	Hard
G3.1	Only the latest published plan shall ever be exposed to participants.	Hard
SG1	The system should improve fairness and consistency of seating map generation.	Soft
SG2	The system should reduce the administrator's manual effort on seating map generation.	Soft
SG3	The system should make allocation results explainable.	Soft
SG4	The system should be usable by administrators without optimization knowledge.	Soft

APPENDIX D Elicitation Questionnaires

Part A: Event scale and context

1. How many participants typically attend PJKIT annual assembly, and how does number of participants vary between events in PJKIT?
2. How many events per year require a formal seating plan, and are the rules the same across them?
3. How long before the event does seat preparation currently start (total days/week needed to generate a seating plan for each event), and how many PJKIT's human forces (staff + volunteer) are involved?

Part B: Venue and layout structure

4. How is the venue physically layout look like? How many rows in total, how many seats are allowed per row?
5. Are there seats that must never be assigned (blocked, reserved, equipment, safety)?

6. Which seats do you consider the "best" seats, and what makes one seat better than another (row, closeness to the center, closeness to the front stage)?

7. Are there seats designated or preferred for participants with mobility difficulties or wheelchairs?

Part C: Tiering (the first-order allocation factor)

8. What is the first factor that decides where a participant will be seated — contribution tiering, contribution amount, participant role (normal public community, staff from PJKIT, monk/sifu from PJKIT, volunteer), seniority, or something else?

9. How are the contribution tiers defined, and what are the boundaries between them such as contribution amount thresholds?

10. Are the number of rows assigned to each contribution tier fixed for every event in PJKIT, or does the number of rows depend on how many people register in each tier? In what order are the tiers seated from the front of the hall, and can a tier's rows ever be shared with the next tier?

11. Do any registrations occupy more than one seat? If so, must the two seats be adjacent, and may they be split across an aisle?

Part D: Ordering, filling, and special placement within the same tier

12. Within the same tier, what decides who sits in front of whom — exact contribution amount, activeness or participation history for event in PJKIT, seniority, or registration order?

13. If two participants have identical standing, how do you break the tie today?

14. When a tier's rows are not completely full, where do the empty seats end up — the back rows of the tier, the outer edges of rows, or spread evenly? Do you fill each row outward from the center aisle?

15. Are there participant categories with special placement expectations (monks, elderly participants, guests of honor), or participants who require an accessible seat?

Part E: Relative importance of the soft factors (weighting)

16. Among the factors just mentioned — contribution-to-seat matching, activeness, category-to-zone suitability — which is the most important to satisfy when they conflict?
17. Could you rank these factors, or say which factor you would sacrifice first if not all can be satisfied?
18. Would you want the ability to change these priorities per event, and who should be allowed to change them?

Part F: Ad-hoc events and dynamic changes

19. What last-minute changes have you experienced in past events (withdrawals, substitutions, walk-in registrations), and how frequently do they happen on the event day itself?
20. When a change happens after the plan is finalized, how much of the existing plan are you willing to change and which participants must not be moved?
21. How is a revised plan being prepared with the current workflow is the mentioned ad hoc event above what happens, and what problems has that caused the event?
22. Who has the authority to approve or revise the seating plan before it is announced to participants?

APPENDIX E Interview Result and Finding

Ref	Elicited Response for the Elicitation Questionnaires
A1	Exact attendance figures are not retained, but attendance across events in the past two years has consistently fallen within the 100–200 participant range.
A2	PJKIT holds one to two events per year that require a formal seating plan. The seating rules are largely stable across events; only the preferred (soft) constraints vary between events.
A3	Seating preparation begins only about two weeks before the event and is carried out by a team of fewer than five senior volunteers who must personally know the participants and understand the significance of each placement; the task cannot be delegated to general staff.
B4	The hall comprises approximately sixteen rows of about sixteen seats, arranged concert style, which a numbered seating section toward the front and a free-seating section toward the back of the venue.
B5	Certain seats are permanently unavailable owing to the building's structural elements; the affected rows simply lose a few seats.
B6	The most desirable seats are those of the east block near the center aisle, while the next most desirable are the west block (西單) seats nearest the aisle.
B7	Participants with limited mobility are seated at the side of the hall for ease of access, as standard practice.
C8	Placement is decided first by contribution tier, then by contribution amount, and thereafter by a combination of seniority and attendance at PJKIT events.
C9	Normally for events in PJKIT, three (3) contribution tiers are recognized, taken directly from the event registration form, which: 1. Emperor tier required for participant to contribute at least RM5000 contribution amount in an event. 2. Merit tier required participants to contribute at least RM3000 contribution amount in an event.

	3. Bodhi tier required participants to contribute at least RM2000 contribution amount in an event.
C10	The number of rows for each contribution tier occupies is not fixed. It depends on the tier's registration count, so a larger Emperor turnout extends further toward the front. Tiers are seated in the order Emperor → Merit → Bodhi from the front, and a row may be shared between two tiers when one tier's registrations end partway along a row, the next tier continues in the same row as to fill up the whole row.
C11	A registration normally occupies two adjacent seats by a married couple, which almost always carry the same name. Registrations larger than two seats are extremely rare and are treated as exceptions.
D12	Within a tier, order is governed first by contribution amount; the remaining factors such as seniority and previous event attendance are combined under weightings that have not yet been finalized from PJKIT side.
D13	Ties are understood to be resolved by registration order (first come, first served), to be confirmed with the client.
D14	Row filling follows the principle already described: rows are packed to capacity outward from the center aisle with no deliberately empty seats, in registration order.
D15	Special categories (monks/sifu, elderly participants, guests of honor) and accessibility needs are accommodated, but manually rather than by an explicit rule.
E16	No single factor is most important; conflicts are resolved pragmatically, tending to satisfy participants more likely to complain. For planning, the model is directed to focus on contribution-to-seat matching and on participant activeness in previous PJKIT events.
E17	A fixed ranking of the soft factors is not possible because the trade-off is situational.
E18	The weightings are not expected to change often, but the event administrator may adjust the emphasis of rules per event to loosen or tighten a given aspect.

F19	Walk-in or late registrations are uncommon because registration closes on a set date. The change that does occur is the absence of a registered participant, sometimes with a replacement taking the vacated seat.
F20	The registration will close after a certain time, but participant absence can still occur after finalization of the seating map generation. When the scenario happens, disruption to already-assigned participants must be kept minimal.
F21	When such a change occurs, the seating map is going to be revised to minimize the effect on already-assigned participants and avoid confusion at the venue.
F22	A plan or a revision is approved by the senior-volunteer working team together with the event administrator (sifu) before it is announced to participants.